Solutions Set

Emergent Causal Order from Entanglement

This is the solutions companion to the homework set associated with the paper *Emergent Causal Order from Entanglement: A Pedagogical Introduction*. Each problem is restated briefly, then a complete solution is given. Numerical values are exact where possible and quoted to four significant figures otherwise.

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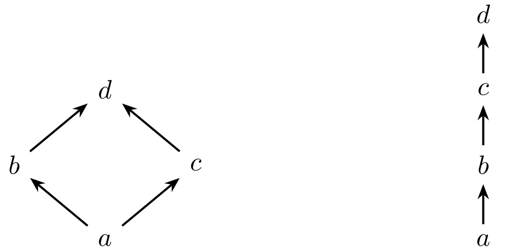
1 Partial orders and Kendall correlation

Problem 1. $S = \{a, b, c, d\}$ with two partial orders defined by their strict relations: $\preceq_1$ has $a \prec b, a \prec c, a \prec d, b \prec d, c \prec d$; $\preceq_2$ has $a \prec b, a \prec c, b \prec c, b \prec d, c \prec d$. Draw the Hasse diagrams, classify all pairs, and compute τ_{12}.

Solution. The Hasse diagrams are:



$\preceq_1$ $\qquad$ $\preceq_2$

For $\preceq_2$, $b \prec d$ given is implied by $b \prec c \prec d$; the Hasse diagram only shows cover edges. The six unordered pairs:

pair	in $\preceq_1$	in $\preceq_2$	class
$\{a,b\}$	$a \prec_1 b$	$a \prec_2 b$	concordant
$\{a,c\}$	$a \prec_1 c$	$a \prec_2 c$	concordant
$\{a,d\}$	$a \prec_1 d$	$a \prec_2 d$	concordant
$\{b,c\}$	incomparable	$b \prec_2 c$	$\preceq_2$ only
$\{b,d\}$	$b \prec_1 d$	$b \prec_2 d$	concordant
$\{c,d\}$	$c \prec_1 d$	$c \prec_2 d$	concordant

So $n_{cc} = 5$, $n_{dd} = 0$, $n_1^o = 0$, $n_2^o = 1$, and $\tau_{12} = 5/(5+0) = 1$. The two orders agree on every pair they both relate; $\preceq_2$ is strictly richer.

Problem 2. Show majorization is transitive on Δ_{d-1}.

Solution. Suppose $x \preceq y$ and $y \preceq z$. By definition, for every $k \in \{1, \ldots, d\}$,

$$\sum_{i=1}^{k} x_i^{\downarrow} \;\le\; \sum_{i=1}^{k} y_i^{\downarrow}, \qquad \sum_{i=1}^{k} y_i^{\downarrow} \;\le\; \sum_{i=1}^{k} z_i^{\downarrow}.$$

Chaining the two inequalities at each k gives $\sum_{i=1}^{k} x_i^{\downarrow} \le \sum_{i=1}^{k} z_i^{\downarrow}$. Since x, y, z all sum to 1, the inequality is automatically an equality at $k = d$. This is exactly the definition of $x \preceq z$.

Problem 3. On $S = \{1,2,3\}$, construct partial orders with (a) $\tau = -1$, (b) τ undefined, (c) $\tau = 0$.

Solution.

(a) $\tau = -1$. Take $\preceq_a$ to be the total order $1 \prec 2 \prec 3$ and $\preceq_b$ to be the reversed total order $3 \prec 2 \prec 1$. Each of the three pairs is comparable in both, but oppositely. $n_{cc} = 0$, $n_{dd} = 3$, $\tau = -3/3 = -1$.

(b) τ undefined. Take $\preceq_a$ relating only $1 \prec_a 2$, and $\preceq_b$ relating only $1 \prec_b 3$. The pairs $\{1,2\}$ and $\{1,3\}$ are each comparable in exactly one order; $\{2,3\}$ is incomparable in both. The both-comparable subset is empty, so the Kendall formula of the paper (eq. 3) has the form $0/0$ and τ is not defined.

(c) $\tau = 0$. Take $\preceq_a$ relating $1 \prec 2$ and $1 \prec 3$ (the bottom is 1, the other two are incomparable). Take $\preceq_b$ relating $1 \prec 2$ and $3 \prec 1$. Then transitivity of $\preceq_b$ gives $3 \prec_b 2$. The three pairs: $\{1,2\}$ is concordant (both $1 \prec 2$); $\{1,3\}$ is discordant ($1 \prec_a 3$ but $3 \prec_b 1$); $\{2,3\}$ is comparable in $\preceq_b$ only. So $n_{cc} = 1$, $n_{dd} = 1$, and $\tau = (1-1)/(1+1) = 0$.

2 Quantum states and the Schmidt decomposition

Problem 4. $|\psi\rangle = \frac{1}{\sqrt{3}}(|00\rangle + |01\rangle + |10\rangle)$. Find Ψ, ρ_A, the Schmidt spectrum, and the entanglement entropy of the cut $\{1\} \,|\, \{2\}$.

Solution. Read off the coefficient matrix Ψ_{ij} where i labels the first qubit and j the second:

$$\Psi \;=\; \frac{1}{\sqrt{3}} \begin{pmatrix} 1 & 1 \\ 1 & 0 \end{pmatrix}.$$

Compute

$$\rho_A \;=\; \Psi\Psi^\dagger \;=\; \frac{1}{3} \begin{pmatrix} 1 & 1 \\ 1 & 0 \end{pmatrix} \begin{pmatrix} 1 & 1 \\ 1 & 0 \end{pmatrix} \;=\; \frac{1}{3} \begin{pmatrix} 2 & 1 \\ 1 & 1 \end{pmatrix}.$$

The trace is 1 as it must be. The determinant is $(2-1)/9 = 1/9$. The eigenvalues solve $\lambda^2 - \lambda + 1/9 = 0$:

$$\lambda_\pm \;=\; \frac{1 \pm \sqrt{1-4/9}}{2} \;=\; \frac{3 \pm \sqrt{5}}{6} \;\approx\; (0.873,\, 0.127).$$

The Schmidt spectrum is therefore $\boldsymbol{\lambda} = (\frac{3+\sqrt{5}}{6},\, \frac{3-\sqrt{5}}{6})$. The entanglement entropy is

$$S = -\lambda_+ \ln \lambda_+ - \lambda_- \ln \lambda_- \;\approx\; -0.873 \cdot (-0.136) - 0.127 \cdot (-2.067) \;\approx\; 0.381 \text{ nats},$$

or equivalently ≈ 0.550 bits.

Problem 5. Show by SVD that $\rho_A = \Psi\Psi^\dagger$ and $\rho_B = \Psi^\dagger\Psi$ have identical nonzero spectra.

Solution. Write $\Psi = USV^\dagger$ with U, V having orthonormal columns and $S = \operatorname{diag}(s_1, \ldots, s_r)$ diagonal with $s_i > 0$. Then

$$\rho_A = \Psi\Psi^\dagger = USV^\dagger VS^\dagger U^\dagger = U(SS^\dagger)U^\dagger, \qquad \rho_B = \Psi^\dagger\Psi = VS^\dagger U^\dagger USV^\dagger = V(S^\dagger S)V^\dagger.$$

Both $SS^\dagger$ and $S^\dagger S$ are diagonal with diagonal entries $(s_1^2, \ldots, s_r^2, 0, \ldots, 0)$ (extended by zeros to the appropriate ambient dimension). Conjugation by an isometry preserves the spectrum, so ρ_A and ρ_B both have spectrum $(s_1^2, \ldots, s_r^2, 0, \ldots, 0)$. In particular their nonzero spectra are identical.

Problem 6. *($\star$)* For $|\Psi\rangle = \cos\theta|000\rangle + \sin\theta|111\rangle$, compute the Schmidt spectrum across the cuts $\{1\} \,|\, \{2,3\}$ and $\{1,2\} \,|\, \{3\}$.

Solution. The state is already in (a tensor-product version of) Schmidt form for both cuts:

$$|\Psi\rangle = \cos\theta\,|0\rangle_1 \otimes |00\rangle_{23} + \sin\theta\,|1\rangle_1 \otimes |11\rangle_{23},$$
$$|\Psi\rangle = \cos\theta\,|00\rangle_{12} \otimes |0\rangle_3 + \sin\theta\,|11\rangle_{12} \otimes |1\rangle_3.$$

The vectors $|0\rangle_1, |1\rangle_1$ are orthonormal; so are $|00\rangle_{23}, |11\rangle_{23}$ and $|00\rangle_{12}, |11\rangle_{12}$ and $|0\rangle_3, |1\rangle_3$. Each expression is therefore already a Schmidt decomposition with coefficients $(\cos\theta, \sin\theta)$. The Schmidt spectrum is $(\cos^2\theta, \sin^2\theta)$ in both cases.

Physically, the GHZ-like state is a single "two-level" bipartite correlation that is shared identically across any non-trivial cut of the chain. No new Schmidt structure appears as more sites move to one side of the cut.

3 Majorization

Problem 7. Decide majorization comparability for four pairs.

Solution.

(a) Cumulative sums of $a = (0.5, 0.4, 0.1)$: $(0.5, 0.9, 1.0)$. Of $b = (0.5, 0.3, 0.2)$: $(0.5, 0.8, 1.0)$. At every k, a's cumulative sum is $\geq b$'s, so $b \preceq a$.

(b) $a = (0.7, 0.2, 0.1)$ has cumsums $(0.7, 0.9, 1.0)$; $b = (0.6, 0.3, 0.1)$ has $(0.6, 0.9, 1.0)$. $a \succeq b$, i.e. $b \preceq a$.

(c) $a = (0.4, 0.3, 0.2, 0.1)$ has cumsums $(0.4, 0.7, 0.9, 1.0)$; $b = (0.4, 0.4, 0.1, 0.1)$ has $(0.4, 0.8, 0.9, 1.0)$. Now b's cumsum dominates, so $a \preceq b$.

(d) $a = (0.5, 0.3, 0.2)$ has cumsums $(0.5, 0.8, 1.0)$; $b = (0.45, 0.40, 0.15)$ has $(0.45, 0.85, 1.0)$. At $k = 1$, $a > b$. At $k = 2$, $a < b$. Neither cumulative sequence dominates the other. a and b are *incomparable* under majorization.

Problem 8. Why can the Bell pair not be deterministically converted into the maximally entangled three-level state via LOCC?

Solution. The Bell pair $\frac{1}{\sqrt{2}}(|00\rangle + |11\rangle)$ has Schmidt spectrum $\boldsymbol{\lambda}_{\text{Bell}} = (\frac{1}{2}, \frac{1}{2})$. Zero-padded to length 3 for comparison: $(\frac{1}{2}, \frac{1}{2}, 0)$. The target state $\frac{1}{\sqrt{3}}(|00\rangle + |11\rangle + |22\rangle)$ has Schmidt spectrum $\boldsymbol{\lambda}_{\text{tgt}} = (\frac{1}{3}, \frac{1}{3}, \frac{1}{3})$.

Theorem 4.4 of the paper says LOCC convertibility $|\psi\rangle \to |\phi\rangle$ requires $\boldsymbol{\lambda}(\psi) \preceq \boldsymbol{\lambda}(\phi)$. Compute cumulative sums:

$$\boldsymbol{\lambda}_{\text{Bell}} : (\tfrac{1}{2},\ 1,\ 1), \qquad \boldsymbol{\lambda}_{\text{tgt}} : (\tfrac{1}{3},\ \tfrac{2}{3},\ 1).$$

At $k = 1$, $\frac{1}{2} > \frac{1}{3}$, so $\boldsymbol{\lambda}_{\text{Bell}}$ does *not* majorization-precede $\boldsymbol{\lambda}_{\text{tgt}}$. Hence no deterministic LOCC protocol exists.

Physically, the target state is more entangled (entropy $\ln 3$) than the Bell pair (entropy $\ln 2$). LOCC cannot increase entanglement, and majorization is precisely the partial order that captures this constraint.

Problem 9. *($\star$)* Show that majorization implies a reverse ordering on Shannon entropy: $x \preceq y \Rightarrow H(x) \geq H(y)$.

Solution. The function $f(p) = -p \ln p$ (with $f(0) := 0$) is concave on $[0, 1]$: indeed $f''(p) = -1/p < 0$ on $(0, 1]$. By the cited Schur-concavity property, the symmetric function $\Phi(x) = \sum_i f(x_i)$ inherits the property that $x \preceq y \Rightarrow \Phi(x) \geq \Phi(y)$ for any concave f. But $\Phi(x) = -\sum_i x_i \ln x_i = H(x)$. Therefore $H(x) \geq H(y)$ whenever $x \preceq y$.

In words: a more spread-out probability vector (lower in majorization) has higher Shannon entropy. Combined with Nielsen's theorem, this gives the standard inequality that LOCC cannot increase entanglement entropy.

4 The lattice Schwinger model

Problem 10. Hopping operator $h_n = X_n X_{n+1} + Y_n Y_{n+1}$. Write as a 4×4 matrix; show its action on the computational basis; explain why it is called "hopping".

Solution. With basis ordering $\{|00\rangle, |01\rangle, |10\rangle, |11\rangle\}$ (first index = first qubit), compute

$$X \otimes X = \begin{pmatrix} 0 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 \\ 0 & 1 & 0 & 0 \\ 1 & 0 & 0 & 0 \end{pmatrix}, \qquad Y \otimes Y = \begin{pmatrix} 0 & 0 & 0 & -1 \\ 0 & 0 & 1 & 0 \\ 0 & 1 & 0 & 0 \\ -1 & 0 & 0 & 0 \end{pmatrix},$$

using $Y|0\rangle = i|1\rangle$ and $Y|1\rangle = -i|0\rangle$ for the second matrix. Adding,

$$h \;=\; \begin{pmatrix} 0 & 0 & 0 & 0 \\ 0 & 0 & 2 & 0 \\ 0 & 2 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{pmatrix}.$$

The action on the basis is therefore

$$h|00\rangle = 0, \qquad h|01\rangle = 2|10\rangle, \qquad h|10\rangle = 2|01\rangle, \qquad h|11\rangle = 0.$$

The operator swaps the two "one-particle" configurations $|01\rangle$ and $|10\rangle$ (with a factor of 2) and annihilates the zero- and two-particle configurations. Under the Jordan–Wigner correspondence $|0\rangle \leftrightarrow$ empty, $|1\rangle \leftrightarrow$ occupied, this is exactly the action of $c_n^\dagger c_{n+1} + c_{n+1}^\dagger c_n$ on the matter field: a single particle hops from one site to its neighbor, with vacuum and doubly-occupied configurations annihilated by the kinetic term.

Problem 11. Compute L_n for $n = 1, 2, 3$ on the all-$|0\rangle$ configuration of an $N = 4$ chain.

Solution. With $L_0 = 0$ and $\langle Z_k \rangle = +1$ on every site, $\frac{1-Z_k}{2} = 0$ for all k, so

$$L_n \;=\; -\sum_{k=1}^{n} \frac{1-(-1)^k}{2}.$$

The summand is 1 for odd k and 0 for even k, so L_n counts (with a sign) the number of odd k in $\{1, \dots, n\}$:

$$L_1 = -1, \qquad L_2 = -1, \qquad L_3 = -2.$$

Physically, the all-$|0\rangle$ configuration is *not* the half-filled staggered vacuum: every odd site is missing its "positron" (the c-number subtraction $\frac{1-(-1)^k}{2}$ encodes the staggered vacuum reference). The cumulative deficit of -1 per odd site to the left of link n is exactly L_n, the net charge to the left of that link. The corresponding electric energy per link, $\frac{g^2 a}{2} L_n^2$, is non-zero, which is why this configuration is not the ground state.

Problem 12. *($\star$)* Show $[M, H] = 0$ for $M = \sum_n Z_n$, where $H = H_{\text{hop}} + H_m + H_E$.

Solution. The pieces H_m and H_E are functions of the Pauli-Z operators only (the latter via the cumulative sums L_n). Pauli-Z operators commute among themselves, so $[Z_k, H_m] = 0$ and $[Z_k, H_E] = 0$ for every k, and therefore $[M, H_m] = [M, H_E] = 0$.

For the hopping term the relevant identity is $[Z_n + Z_{n+1},\, X_n X_{n+1} + Y_n Y_{n+1}] = 0$. Using $[Z_n, X_n] = 2iY_n$ and $[Z_n, Y_n] = -2iX_n$ (the basic Pauli commutators) and $[A, BC] = [A, B]C + B[A, C]$:

$$\begin{aligned}
[Z_n,\, X_n X_{n+1}] &= 2iY_n X_{n+1}, & \qquad [Z_n,\, Y_n Y_{n+1}] &= -2iX_n Y_{n+1}, \\
[Z_{n+1},\, X_n X_{n+1}] &= 2iX_n Y_{n+1}, & \qquad [Z_{n+1},\, Y_n Y_{n+1}] &= -2iY_n X_{n+1}.
\end{aligned}$$

Adding the first column and the second column separately and then combining,

$$[Z_n + Z_{n+1},\, X_n X_{n+1} + Y_n Y_{n+1}] = (2iY_n X_{n+1} - 2iX_n Y_{n+1}) + (2iX_n Y_{n+1} - 2iY_n X_{n+1}) = 0.$$

Operators on disjoint sites commute, so Z_k for any $k \notin \{n, n+1\}$ commutes with $X_n X_{n+1} + Y_n Y_{n+1}$ trivially. Therefore

$$[M, H_{\text{hop}}] \;=\; \frac{1}{4a} \sum_{n=1}^{N-1} [\sum_k Z_k,\, X_n X_{n+1} + Y_n Y_{n+1}] \;=\; 0.$$

Combining: $[M, H] = 0$. Translated to the fermion picture, this says the total fermion number $\sum_n c_n^\dagger c_n = (N - M)/2$ is conserved by the dynamics.

5 Lieb–Robinson and causal orders

Problem 13. Five-site chain, snapshots at $t \in \{0, 0.5, 1.0, 1.5\}$. Count LR-related space-time pairs at $v = 0.5$ and at $v = 2$.

Solution. The condition for $(x_a, t_a) \preceq_{\text{LR}} (x_b, t_b)$ is $t_a < t_b$ and $|x_a - x_b| \le v\,(t_b - t_a)$. Time differences in the snapshot set come from $\Delta t \in \{0.5, 1.0, 1.5\}$ with multiplicities 3, 2, 1 respectively (a standard count of ordered pairs $\binom{4}{2} = 6$). Spatial gaps on a five-site chain are $0, 1, 2, 3, 4$, with multiplicities $5, 4, 3, 2, 1$.

(a) $v = 0.5$**.** At $\Delta t = 0.5, 1.0, 1.5$ the maximum admissible spatial gap is $0.25, 0.5, 0.75$ respectively — below 1 in every case. Only same-site (gap $= 0$) pairs are LR-related:

$$\#\{\text{pairs}\} \;=\; (3 + 2 + 1) \times 5 \;=\; 30.$$

(b) $v = 2$. At $\Delta t = 0.5$, gap ≤ 1 admits $5 + 4 = 9$ spatial pairs; at $\Delta t = 1.0$, gap ≤ 2 admits $5 + 4 + 3 = 12$; at $\Delta t = 1.5$, gap ≤ 3 admits $5 + 4 + 3 + 2 = 14$. Total:

$$3 \cdot 9 \; + \; 2 \cdot 12 \; + \; 1 \cdot 14 \; = \; 27 + 24 + 14 \; = \; 65.$$

A wider LR cone admits more than twice as many pairs as the narrower one.

Problem 14. Show $\preceq_{\mathrm{LR}} \subseteq \preceq_{\mathrm{cs}}$ on the regular chain, hence $\tau(\mathrm{LR}, \mathrm{cs}) = +1$.

Solution. On the regular 1+1D chain $\preceq_{\mathrm{cs}}$ is the time-only order: $(A, s) \preceq_{\mathrm{cs}} (B, t)$ iff $s < t$. Now suppose $(A, s) \preceq_{\mathrm{LR}} (B, t)$. By the definition of $\preceq_{\mathrm{LR}}$ this requires $s < t$ (and a spatial-gap condition that we do not need here). Hence $(A, s) \preceq_{\mathrm{cs}} (B, t)$. This gives the inclusion $\preceq_{\mathrm{LR}} \subseteq \preceq_{\mathrm{cs}}$ as relations on the same label set.

Consequently every pair comparable in both orders relates the same way (because $\preceq_{\mathrm{LR}}$ is just a sub-relation of $\preceq_{\mathrm{cs}}$), so $n_{cc} = |\preceq_{\mathrm{LR}}|$, $n_{dd} = 0$, and $\tau(\mathrm{LR}, \mathrm{cs}) = (|\preceq_{\mathrm{LR}}| - 0)/(|\preceq_{\mathrm{LR}}| + 0) = +1$. This is the trivial-agreement caveat: the Kendall coefficient is forced to its extremal value by the lattice geometry, not by any property of the state.

6 Comparison statistics

Problem 15. Compute $n_{cc}, n_{dd}, n_a^o, n_b^o, \tau$ for the two specified partial orders on $\{1, 2, 3, 4, 5\}$.

Solution. Transitive closures:

- $\preceq_a$: $1 \prec 2$, $1 \prec 3$, $2 \prec 3$, $4 \prec 5$.
- $\preceq_b$: $1 \prec 3$, $2 \prec 4$, $2 \prec 5$, $4 \prec 5$.

The ten unordered pairs:

pair	in $\preceq_a$	in $\preceq_b$	class
$\{1, 2\}$	$1 \prec_a 2$	incomparable	$\preceq_a$ only
$\{1, 3\}$	$1 \prec_a 3$	$1 \prec_b 3$	concordant
$\{1, 4\}$	incomparable	incomparable	both incomparable
$\{1, 5\}$	incomparable	incomparable	both incomparable
$\{2, 3\}$	$2 \prec_a 3$	incomparable	$\preceq_a$ only
$\{2, 4\}$	incomparable	$2 \prec_b 4$	$\preceq_b$ only
$\{2, 5\}$	incomparable	$2 \prec_b 5$	$\preceq_b$ only
$\{3, 4\}$	incomparable	incomparable	both incomparable
$\{3, 5\}$	incomparable	incomparable	both incomparable
$\{4, 5\}$	$4 \prec_a 5$	$4 \prec_b 5$	concordant

Counts: $n_{cc} = 2$, $n_{dd} = 0$, $n_a^o = 2$, $n_b^o = 2$. Therefore

$$\tau_{ab} \; = \; \frac{2 - 0}{2 + 0} \; = \; +1.$$

Note the contrast with raw pair counts: only 4 of the 10 pairs are comparable in both orders, but on those 4 they agree unanimously. This illustrates why the asymmetric counts n_a^o and n_b^o carry information beyond τ: they tell you *how much overlap* there is to compare in the first place.

7 Interpretation

Problem 16. State the difference between the three falsification criteria and give one example of each.

Solution. **Strong falsification (criterion 1).** $\preceq_{\text{maj}}$ relates a pair whose endpoints lie outside the LR cone at the corresponding time difference. Concretely: there exist labels $(A, s), (B, t)$ with $\boldsymbol{\lambda}_A(s) \preceq \boldsymbol{\lambda}_B(t)$ but $\text{gap}(A, B) > v_{\text{LR}}\,(t - s)$. Such a pair would require a faster-than-Lieb–Robinson information channel to be consistent with the hypothesis, so it directly refutes (H).

Example. In Phase 5 (Table 2 of the paper), the count $n_{\text{maj}\notin\text{LR}}$ at $v = 1$ is positive in every regime; each contributing pair is a strong-falsification observation.

Weak falsification (criterion 2). $\preceq_{\text{maj}}$ disagrees with a non-trivially-defined $\preceq_{\text{cs}}$ on a non-vanishing fraction of comparable pairs. This is meaningful only when $\preceq_{\text{cs}}$ has within-time-slice structure — i.e., on a non-regular causet — and asks whether the prescribed causet structure controls the entanglement order.

Example. On a 1+1D causal dynamical triangulation with multi-vertex antichains, find a pair related by $\preceq_{\text{cs}}$ but not $\preceq_{\text{maj}}$, where the cs relation is genuinely within-slice (not implied by the time order). This is the Phase 6 test of the implementation plan.

Trivial agreement (criterion 3). $\preceq_{\text{cs}}$ is forced to coincide with $\preceq_{\text{LR}}$ by lattice regularity. Any agreement between $\preceq_{\text{maj}}$ and $\preceq_{\text{cs}}$ is then uninformative because $\preceq_{\text{cs}}$ has no extra structure to confirm against.

Example. On the regular chain, $\preceq_{\text{cs}}$ reduces to the time-only order and contains $\preceq_{\text{LR}}$ as a strict subset. An observation $\tau(\text{maj}, \text{cs}) \approx +1$ in this geometry would *not* support (H), because $\preceq_{\text{cs}}$ is here a coarse total-time order that any half-decent state-derived ordering will mostly agree with.

Strong but not weak. A pair on the regular chain whose spatial gap exceeds $v_{\text{LR}}(t - s)$ and is $\preceq_{\text{maj}}$-related is a strong-falsification observation. It is not a weak-falsification observation because the regular chain has trivial $\preceq_{\text{cs}}$ structure — criterion 2 is not even posable in this geometry.

Problem 17. *($\star$)* Why is the trend $\tau(\text{maj}, \text{LR})\downarrow$ as $v\uparrow$ counterintuitive, and what does it imply?

Solution.

(a) *Naive expectation.* Widening v admits more pairs into $\preceq_{\text{LR}}$. If those new pairs were drawn at random from the $\preceq_{\text{maj}}$-comparable set with the same agreement rate as the existing pairs, the both-comparable count $n_{cc} + n_{dd}$ would grow and τ would stay roughly constant or, if the new pairs were even better aligned with $\preceq_{\text{maj}}$ than the old ones, rise. The intuition that "a bigger cone = more causal pairs = closer to capturing $\preceq_{\text{maj}}$" predicts $\tau\uparrow$.

(b) *What we see.* τ drifts *down.* The pairs newly admitted by widening the cone do not merely fail to improve the agreement; they actively pull τ down by being net discordant with $\preceq_{\text{maj}}$ on the both-comparable subset (recall pairs that are LR-comparable but not maj-comparable sit in n_b^o and do not affect τ directly). So at least some of the newly-admitted "causal" pairs are ordered by $\preceq_{\text{maj}}$ in the opposite direction from the time order.

(c) *Picture.* Imagine $\preceq_{\text{maj}}$ as carrying its own effective "cone" of slope v_{maj}. Pairs strictly inside that cone tend to be concordant with the time direction. Pairs in the LR ring outside v_{maj} but inside v contribute mostly discordant relations. As v grows past v_{maj}, the discordant ring grows faster than the concordant interior, so τ falls.

Problem 18. *($\star\star$)* Suggest two refinements of (H) that could survive the Phase 5 result.

Solution. There is no unique correct answer. Two natural refinements:

Refinement 1: bounded-length cuts. Restrict the cut family $\mathcal{F}$ to intervals of length $\leq L_{\max}$. Hypothesis (H_1): the three orders coincide when restricted to short-interval cuts.

Test: re-run the agreement scan at varying $L_{\max}$. Support: a sharp rise in $\tau(\mathrm{maj}, \mathrm{LR})$ as $L_{\max}$ shrinks would suggest the discordance is concentrated in long-interval cuts, perhaps because long intervals see boundary effects or accumulate local truncation errors.

Refinement 2: charge-sectorized majorization. Compute Schmidt spectra restricted to fixed total-charge sectors of each bipartition — i.e., for each region A and each value q of $\sum_{n\in A} \frac{1-Z_n}{2}$, build the Schmidt spectrum $\boldsymbol{\lambda}_A^{(q)}(t)$ of the projection onto the q-sector. Define $\preceq_{\mathrm{maj}}^{(q)}$ accordingly. Hypothesis (H_2): the sectorized maj order $\preceq_{\mathrm{maj}}^{(q)}$ coincides with $\preceq_{\mathrm{LR}}$ on each sector. Test: re-run with quantum-number-resolved spectra, already supported by the underlying ITensor implementation. Support: the global mixing of charge sectors might be the source of the $\sim 50\%$ overflow; resolving it could push the overflow fraction to small.

Other plausible refinements include: time-coarse-graining (group consecutive snapshots), substituting Hasse-graph edit distance for exact-equality of orders, and weighted Kendall coefficients that discount pairs near the cone boundary.

8 Computational exercises (optional)

Problem 19. Run `lightcone_vs_majorization.py` and verify Table 1.

Solution. The expected wall-clock time on a single laptop CPU is roughly 80 seconds. Output to within $\sim 10^{-4}$ of the paper's Table 1 is expected; small discrepancies arise from compiler-level numerical differences (LAPACK / BLAS implementation, FMA contraction) and Krylov-subspace basis ordering and are not physically significant. A run that disagrees by more than $\sim 10^{-3}$ on any τ value indicates a configuration mismatch (commonly: bond dimension or DMRG sweep count) and warrants investigation.

Problem 20. *($\star$)* Add an intermediate $m/g = 1.0$ regime and compare.

Solution. Adding a regime is a one-line modification to `lightcone_vs_majorization.py`: add a fourth call to `scan_vlr` with `m_over_g=1.0` between regimes A and B.

The expected qualitative finding is monotone behavior: the intermediate $m/g = 1.0$ produces $\tau(\mathrm{maj}, \mathrm{LR})$ values *between* the light-quark ($m/g = 0.5$) and heavy-quark ($m/g = 5.0$) values at every v, with τ rising smoothly with m/g. The physical intuition is that increasing m/g stiffens the flux tube and suppresses pair production; the entanglement spreading slows; the Schmidt spectra evolve more in time than across cuts; $\preceq_{\mathrm{maj}}$ tracks the time order more closely; and $\tau(\mathrm{maj}, \mathrm{LR})$ rises.

A non-monotone observation would be surprising and physically interesting — it would imply a non-trivial competition between two effects (e.g. string fluctuations vs. pair production) that maximally disorganizes the spectra at intermediate m/g.